

Coherence-resolution admissibility: deriving the mode-count cutoff from the dressed propagator (gate STEP-5B, hypothesis H-ADM)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The AddA chain reduced STEP-5B closure to the named hypotheses H-KBAL (lifted in principle, AddA v1.2) and H-ADM ($n \leq n_{\text{adm}}$ with the verified threshold $n_{\text{adm}} \lesssim 1.59 \times 10^5$ at the anchor). This note derives a CANDIDATE value of n_{adm} from TECT microphysics — the coherence resolution of the dressed propagator — rather than postulating it. The result is registered at T3 (PROOF SKETCH): it is a physically derived class-definition amendment proposal, and its promotion to a binding admissibility definition requires operator sign-off. No tier action is claimed.

2. Derivation

The dressed quadratic kernel at the production anchor is $D(k) = \hat{r} + C(k^2 - q_0^2)^2$ with $\hat{r} = r_R + 2\lambda'I$. At $I = 4 \times 10^{-4}$: $\hat{r} = 0.3110$ and

$$\frac{\hat{r}}{Cq_0^4} = 1.45,$$

the STRONGLY DRESSED regime: the propagator is barely shell-peaked. Expanding about the shell, $D \approx \hat{r} + 4Cq_0^2(k - q_0)^2$, the correlation length and the induced angular resolution are

$$\xi = 2q_0\sqrt{C/\hat{r}} = 2.44, \quad \theta_{\min} = \frac{1}{q_0\xi} = \frac{\sqrt{\hat{r}}}{2q_0^2\sqrt{C}} = 0.603 \text{ rad.}$$

Two shell modes separated by an angle $\theta < \theta_{\min}$ produce a beat wavevector $q_0\theta < 1/\xi$ whose phase cannot be resolved within one correlation volume: within the Gaussian–Hartree variational class they are not independent coherent READINGS but a single reading plus sea-type fluctuation, which the layer (hypothesis H-LAYER) already accounts for. Cap packing on the shell then caps the independent-reading count:

$$n_{\text{adm}} \simeq c_{\text{geo}} \frac{4\pi}{\theta_{\min}^2} = c_{\text{geo}} (q_0\xi)^2 4\pi/(4\pi) \cdot \dots \approx 35 \quad (c_{\text{geo}} = 1); \quad \leq 140 \quad (\times 4 \text{ conservative}).$$

The cutoff is nearly intensity-independent (35/34/32 at $I = 4 \times 10^{-4}/10^{-3}/2 \times 10^{-3}$) because $\hat{r}$ varies mildly across the anchor band.

3. Closure consequence

With the OFFICIAL theorem-grade energy bound (AddA, $K(n) \leq 8 + 4\sqrt{14}\kappa^4\sqrt{n}$) evaluated at the conservative $4n_{\text{adm}}$:

$$K(4n_{\text{adm}}) \approx 184 < K_{\text{budget}}(I) = 5972/927/221 \quad \text{at } I = 4 \times 10^{-4}/10^{-3}/2 \times 10^{-3},$$

margin ratios $\times 32.4 / \times 5.1 / \times 1.2$. STEP-5B therefore CLOSES at all anchor intensities under H-ADM-COH, with the $I = 2 \times 10^{-3}$ corner THIN ($\times 1.2$) under the $\times 4$ -conservative packing ($\times 2.4$ at $c_{\text{geo}} = 1$); the de-thinning (exact cap-packing constant; replacing the heuristic $\theta_{\min}$ by a linewidth-integral criterion) is registered as the follow-up refinement.

4. Status and the amendment proposal

The argument above is a PROOF SKETCH (T3): the coherence-resolution criterion is physically standard (mode separation below the inverse correlation scale is sea-like), and it quantifies the operator’s verdict-#9 observation that arbitrarily dense shell peaks should reclassify into the Reading-H sea or a continuum angular distribution. What requires operator decision is the CLASS-DEFINITION AMENDMENT:

- **H-ADM-COH (proposed)**: an admissible competitor pattern is a finite set of shell readings with pairwise angular separation $\geq \theta_{\min}(\hat{r})$; mass at finer angular scales belongs to the sea sector of the variational layer.

Under sign-off, STEP-5B closes (at the stated margins) for the entire amended class; without it, H-ADM remains a named hypothesis with the derived candidate value.

5. Numerical verification

Script [codes/vacuum/beyond_layer_gershgorin_bound.py](#) (v1.11.0; 175/175 self-test asserts PASS; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#)).

Quantity	Value	Check
$\hat{r}/(Cq_0^4)$ at anchor	1.45	strongly dressed regime
$\xi, q_0\xi, \theta_{\min}$	2.44, 1.66, 0.603 rad	parabolic expansion of D
n_{adm} ($c_{\text{geo}}=1 / \times 4$)	35 / 140	nearly I -independent (35/34/32)
$K(4n_{\text{adm}})$ vs budget	184 vs 5972/927/221	ratios $\times 32.4 / \times 5.1 / \times 1.2$
runtime discipline	suite 24.5 s (< 45 s cap)	hotspots vectorized (v1.11.0)

Quantitative sanity check (mandatory): $\theta_{\min} = \sqrt{\hat{r}}/(2q_0^2\sqrt{C}) = 0.5577/(2 \cdot 0.4626) = 0.603$ — dimensionally an angle ($\text{length}^{-1}/\text{length}^{-1}$); $n_{\text{adm}}\theta_{\min}^2 = 4\pi$ recovers the sphere solid angle; the closure inequality fails if n_{adm} were 1.59×10^5 -scale, confirming the cutoff (not the budget) is the binding element.

6. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “The criterion is heuristic: beats below $1/\xi$ are suppressed, not forbidden.” VALID-with-mitigation: this is why the note is T3 and an AMENDMENT PROPOSAL, not a theorem.

The rigorous form (free-energy indistinguishability: sub-resolution mode splittings change the variational energy by less than the sea-sector accounting) is the registered follow-up; the $\times 4$ packing band absorbs criterion ambiguity at the production anchor ($\times 32$ margin).

- (β) “ ξ uses the parabolic expansion, but the shell is broad ($\hat{r} \sim Cq_0^4$).” VALID-with-mitigation: the broad shell weakens the parabolic approximation radially, but the angular-resolution argument uses only the transverse beat scale $1/\xi$, and a broader shell can only SHORTEN the true coherence range (more soft modes to dephase against), making the quoted n_{adm} conservative in the relevant direction. An exact linewidth-integral criterion is registered as the refinement, and the I -insensitivity of the result (35/34/32) indicates robustness.
- (γ) “Circular dependence: $\hat{r}$ contains $\lambda'I$, the quantity under study.” DISMISSED: $\hat{r}$ is the DRESSED kernel fixed by the anchor calibration (Math434 chain); the cutoff is evaluated at each anchor intensity separately and varies by $< 10\%$ across the band.
- (δ) “The $I = 2 \times 10^{-3}$ margin ($\times 1.2$) is too thin to quote closure.” VALID-with-mitigation: quoted honestly as THIN; at $c_{\text{geo}} = 1$ the ratio is $\times 2.4$; the de-thinning refinement is registered, and the production anchor itself has $\times 32$.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / AddB (serving STEP-5B)
Precise statement:	H-ADM-COH derivation (PROOF SKETCH, T3): $\xi = 2 q_0 \sqrt{C/r_{\text{hat}}} = 2.44$; $\theta_{\text{min}} = 1/(q_0 \xi) = 0.603$ rad; $n_{\text{adm}} \sim 35$ ($\times 4$: 140), nearly I -independent. Consequence: $K(4 n_{\text{adm}}) = 184 < \text{budget at ALL anchor intensities } (\times 32.4 / \times 5.1 / \times 1.2) \Rightarrow \text{STEP-5B closes under the proposed class amendment H-ADM-COH.}$
Scope:	Class-definition amendment PROPOSAL; binding status requires operator sign-off. Margins quoted with the $\times 4$ -conservative packing band; $I = 2e-3$ corner THIN ($\times 1.2$).
Dependencies:	AddA v1.2 (official $\sqrt{n}$ route + H-KBAL lift); B5 main chain v2.0; anchor calibration (Math434 chain via archive).
Evidence grade:	PROOF SKETCH (T3) + EXECUTED (all numbers machine-verified)
Reproduction command:	<code>python codes/vacuum/beyond_layer_gershgorin_bound.py</code>
Expected output:	claims 175/175 PASS; result.json under <code>claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/</code>
Falsification gate:	An admissible (resolution-respecting) pattern with $n > 4 n_{\text{adm}}$ at any anchor intensity; or a demonstration that sub-resolution splittings change the variational energy beyond the sea-sector accounting (would refute the criterion, not the theorems).
Tier before / after:	T3 (this note) / B5 unchanged at T4+

No-overclaim statement: The cutoff derivation is a PROOF SKETCH and a PROPOSED amendment - NOT a binding closure. STEP-5B remains formally OPEN until operator sign-off on H-ADM-COH (or DR-2 closes the unconditional route). No tier action on B1/B2; Reading-H stays T5.

Next required action: Operator decision on H-ADM-COH; de-thinning of the $I = 2e-3$ corner; rigorous indistinguishability lemma; DR-2 remains the unconditional sharp route.